

REPORT ON RUBBER IN TRIPURA AS ON 31ST MARCH, 2023

1. Status of Plantation & Production

1.1 Prelude

Rubber cultivation dates back to 1963 when State Forest Dept. had introduced this crop to Tripura aiming at soil conservation measures with the help of Rubber Board. Majority of rubber trees is of RRIM 600 clone. Average stand per ha in mature area is less than 280 which is primarily caused by tempests/cyclones that happen almost every year. Insurgency, which overran normalcy in the state for almost 2 decades during 80s and 90s, also had devastating effects on plant health and its population. The state is blessed with less disease-pest attacks. Annual wintering is yet another external limiting factor.

1.2 Demography

Rubber plantations in Tripura have 3 types of ownerships. TFDPC Ltd has about 7,018 ha rubber plantations in denuded forest areas. In estate sector there are 4 entities to have a total area of 347 ha (Murticherra Tea Estate- 90 ha, Manu Valley Tea Estate- 75 ha, Binodini Tea Estate- 57 ha, Stevia Realtors Pvt. Ltd.- 125 ha). Rest of the areas covering 89,807 ha belongs to individual small growers of the state. TFDPC Ltd., TRPC Ltd. and TTAADC/Tribal development departments have developed 2,120 ha, 10,852 ha and about 6,315 ha respectively for individual beneficiaries. So, balance area, which is about 70,807 ha has been arranged by individual rubber growers themselves with the support of Rubber Board's Rubber Development Schemes that even includes 3800 ha, developed under Board's prestigious Block Plantation Project. There are about 1.26 lakh individual rubber growers most of whom hail from rural Tripura. It is estimated that 82% area under rubber is owned by unprivileged sections of the society (ST & SC categories). Today the vast majority of the rubber plantations in Tripura pertain to small growers. Average holding size is assessed as 0.75 ha.

Ownership Patterns				
SI	Entities	Forest/Estate	Individual Beneficiary oriented	Total
1	TFDPC	7,018	2,120	9,138
2	Estates	347	0	347
3	TRPC	0	10,852	10,852
4	ADC/TDD	0	6,315	6,315
5	Rubber Board	0	70,519	70,519
Total		7,365	89,807	97,172

1.3 Rubber Development Agencies

Extension of rubber plantation took momentum once Rubber Board opened up one one-man-office at Agartala in 1967. Rubber Board played a pivotal role in domesticating rubber outside govt. owned estate sector with financial assistances what were payable through Board's RPD Scheme. In 1976 Tripura Forest Development & Plantation Corporation (TFDPC) Ltd. was set up to get a big leap for expansion of rubber mostly in denuded forest land. TFDPC Ltd. is the largest rubber estate in the country. The one-man-office at Agartala was upgraded to the status of Regional Office during 1979 for undertaking further development and extension activities. A Regional Research Station aiming at location specific R&D was also established in 1979, with a farm of 85 ha at Taranagar. Laying special emphasis for the rehabilitation of the "Jhumia" tribal, Tripura Rehabilitation Plantation Corporation (TRPC) Ltd. was established in 1983. Subsequently, Nucleus Rubber Estate & Training Centre (NRETC), under a Joint Rubber Production Commissioner, was established in 1985 within the scheme of "Accelerated Development of Rubber in Tripura". NRETC soon started functioning as the Zonal Office (ZO) for the state of Tripura. Until 80s, despite all efforts made, the level of adoption of rubber was poor, mainly because of grower's scepticism and lack of a matching approach. It was, at this juncture, NRETC had intervened and educated the stakeholders in rubber culture. Under NRETC, a nursery cum demonstration plot was established at Tulakona with an area of 14.32 ha which has propagated high yielding improved clones for the state since 1986. At present it needs complete renovation.

NRETC/ZO has also been implementing Block Plantation Scheme since 1992. Under this scheme a compact land (Block) is identified where mainly ST families have ownership over that land. Block plantations were raised under the direct technical supervision of the Board's officials. The beneficiaries were also supported with continuous wage employment for doing different up-keep operations throughout the immaturity period of their plantations. On attaining tappability, the beneficiaries were imparted training in tapping, processing and marketing procedures under the umbrella of a Rubber Producers' Society. They were given infrastructure like Group Processing Centres equipped with smoke houses and utensils, sheeting rollers, community halls, effluent treatment plants etc. Rubber Board also organized rubber growers of the state under the umbrella of Rubber Producers' Societies (RPSs). There are 234 nos. of Rubber Producers' Societies (RPSs) which cover less than 20% individual growers.

Five new Regional Offices at Udaipur, Dharmanagar, Ambasa, Bishramganj and Santirbazar had been established under NRETC/Zonal Office in 1988, 1994, 2010, 2014 and 2016 respectively. Twenty-five Field Stations at various strategic locations (from Kanchanpur in North to Manubazar in South) are also established to cater the need of the rubber growers. Two Tappers' Training Schools at Juri (North Tripura) and Amtali (West Tripura) restlessly generated manpower for tapping and primary processing of rubber latex. Subsequently TTS at Amtali was abandoned. Tripura Tribal Areas Autonomous District Council (TTAADC) and State's Tribal Welfare Departments have also added their efforts in spreading rubber from one village to another.

1.4 Role of Rubber in Socio-economic up-liftment

Rubber has brought out a paradigm shift in socio-economic panorama in Tripura. It is now the most socially accepted and commercially viable plantation crop in the state. It has reinforced the economic value of plantation activity with its social significance. Rubber in Tripura, apart from its commercial value, has also been playing an imperative role in rural development. It delivers such socio-economic activities around which the rehabilitation projects for shifting cultivators had once been brought up.

Land use pattern also got redefined. It is told that this crop has once helped in reducing insurgency in the state.

The main attractions for rubber in Tripura have been

1. No matching income from same type of uplands/terrain,
2. About 30 year's continuous income after 7 years of gestation period,
3. Rubber yielding on every alternative day,
4. Rubber marketing is the easiest one compared to any other crop.

1.5 Area & Production

Total area under rubber in Tripura as on 31st March, 2023 is estimated to be 97,171.93 ha out of which 78,084.13 ha is in yielding stage. Production of rubber has been estimated to be 1,02,989.36 mt. Most of the mature area is under tapping. The state stands second in the country in terms of area and production of natural rubber.

1.6 District-wise Distribution as on 31st March, 2023

Area & Production of Rubber in Tripura 2022-23						
Sl	Name of the District	Immature (ha)	Mature (ha)	Total Area (ha)	Production (mt)	Productivity kg/ha
1	North Tripura	2,096.92	8,708.40	10,805.32	10,336.87	1,187.00
2	Unakoti	1,190.88	3,454.04	4,644.92	4,072.31	1,179.00
3	Dhalai	3,894.57	5,112.48	9,007.05	5,950.92	1,164.00
4	Khowai	390.84	5,134.88	5,525.72	6,526.43	1,271.00
5	West Tripura	3,388.20	13,684.86	17,073.06	17,585.05	1,285.00
6	Sepahijala	2,325.84	16,569.18	18,895.02	23,229.99	1,402.00
7	Gomati	2,072.60	11,480.81	13,553.41	18,908.90	1,647.00
8	South Tripura	3,727.95	13,939.48	17,667.43	16,378.89	1,175.00
Total		19,087.80	78,084.13	97,171.93	1,02,989.36	1,318.95

Sepahijala district tops in the list with the highest area under rubber with 18,895.02 ha (19.44%) while Unakoti district has the lowest figure of 4,644.92 ha (4.78%). Sepahijala district also stands highest in production (23,229.99 mt) with 22.56% share while Unakoti district has the lowest production of 4,072.31 mt (3.95%) share.

1.7 New & Re-planting Initiatives

Trends of replanting program had been very meagre since beginning against potential figures of about 2,500 ha/year. As per available report there were only 492.30 ha re-planting last year, that too, exclusively by TFDPC Ltd. Absence of rubber wood processing facility could not ensure remunerative price for rubber trees on felling.

Recent INROAD initiative, sponsored by Automotive Tyre Manufacturers' Association (ATMA) and implemented by the Rubber Board for new planting/re-planting on 30,000 ha over a period of next 5 years commencing from 2021 has opened up novel prospects of planting rubber in Tripura once again. This project has been named as 'Chief Minister's Rubber Mission' in Tripura. Within first two years, Tripura has raised 10,773 ha under INROAD scheme.

INROAD Plantation (Chief Minister's Rubber Mission in Tripura) Total Planting in 2022									
Sl	Name of the District	RB	TRP PTG	TFDPC	TRPC	ADC	BDO	SDWO	Total
		(ha)	(ha)	(ha)	(ha)	(ha)	(ha)	(ha)	(ha)
1	West Tripura	140.48			31.00	239.80			411.28
2	Dhalai	808.27	655.54		139.49	404.53	87.00	359.00	2453.83
3	North Tripura	600.70	85.00	85.63	92.00	39.00			902.33
4	Unakoti	376.95	50.00	76.67	162.17	108.00			773.79
5	Khowai	250.00	10.00		433.08	216.50			909.58
6	South Tripura	545.00		249.00		32.00			826.00
7	Sepahijala	170.51		81.00	231.71	173.92			657.14
8	Gomati	848.05	255.00		363.00				1466.05
Total		3739.96	1055.54	492.30	1452.45	1213.75	87.00	359.00	8400.00

INROAD Plantation (Chief Minister's Rubber Mission in Tripura) New Planting in 2022									
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1	West Tripura	140.48			31.00	239.80			411.28
2	Dhalai	808.27	655.54		139.49	404.53	87.00	359.00	2453.83
3	North Tripura	600.70	85.00		92.00	39.00			816.70
4	Unakoti	376.95	50.00		162.17	108.00			697.12
5	Khowai	250.00	10.00		433.08	216.50			909.58
6	South Tripura	545.00				32.00			577.00
7	Sepahijala	170.51			231.71	173.92			576.14
8	Gomati	848.05	255.00		363.00				1466.05
Total		3739.96	1055.54		1452.45	1213.75	87.00	359.00	7907.70

2. Harvesting Procedures

2.1 Tappers

There are about 40% plantation in the state which is tapped by hired tapper while rest 60% by self-tapping by the growers or their family members. Tapping task remains around 250. Skilling & up-skilling of tappers is a continuous endeavour of Rubber Board. Tappers mostly get remuneration on monthly basis. Common tapping intensity is alternate day tapping (s/2 d/2).

2.2 HRD of Plantation Workers

The Tappers Training School at Juri under RO, Dharmanagar is imparting training to 10 nos. of tappers per batch.

Apart from this, National Institute for Rubber Training (NIRT), Rubber Board sponsored programs have developed harvesting and up-keep skill among more than 3000 number of harvesters, processors & other stakeholders.

2.3 Productivity

Average productivity remains around 1200-1300 kg/ha/year during last couple of years as constant up-skilling and sporadic productivity enhancing programs are scaling up production in one hand but old plantations gradually grow senile to suppress production capacity on the other. Average productivity during 2022-23 has been 1319 kg/ha.

2.4 Productivity Enhancement Measures

Attitude for adoption of GAPs is not mention worthy. Productivity enhancing measures like manuring, soil-water conservation, rain-guarding are not common in practice.

There are stray cases of silt pit making in about 400 ha. Manuring has been estimated to be done in about 10,000 ha. Rain-guarding has been in about 3,000 ha with active involvement of Board's Regional Offices and promoted company, Manimalayar Rubbers.

3. Processing Modules

3.1 Primary Processing facilities

The primary latex processing facilities available in Tripura include Sheet rubber, ISNR & Cenex making units.

Marketable forms of natural rubber in Tripura include Preserved Field Latex (PFL), Field Coagulum (FC), Sheet rubber (RSS & ADS grades), Indian Standard Natural Rubber (ISNR), Cenex and Creep rubbers.

3.2 Quality Up-gradation Initiative

Quality of produce has been a critical issue in the state until middle of 2010s. A study, conducted by Regional Research Station, Agartala had found 93% sheet rubber as ungraded during 2012 (*Sharma, Gaurav; George K., Tharian; Veeraputhran, S.; Joseph Joy and Dey, S.K. (2012) Marketing Efficiency of Organized and Unorganized Rubber Growers in Tripura: A Comparative Analysis. In: International Rubber Conference, held at Rubber Research Institute of India, Kerala, October 29- 31).*

Board's promoted Manimalayar Rubbers Pvt. Ltd. has brought out paradigm shift in the attention and attitude of the producers' mind-set and in the arrangement and attire in their processing units for producing RSS and tyre grades of sheet rubber. This has been arranged by introducing 1-day practical training (Grade Rubber Orientation Workshop-GROW), renovations of ineffective smoking units, development of new generation semi mechanized latex processing model (Integrated RSS Making Units-IRMUs) and through grade-wise price fixing formula. Such endeavours could attract many from individual, estate, RPS and PSU sectors. A remarkable increase in the shares of RSS grades (7% in 2012 and 70% as of now) against off-grade sheet has been achieved by such initiatives. Manimalayar Rubbers Pvt. Ltd. also had made Automotive Tyre Manufacturers' association (ATMA) delegates visit Tripura in 1997, convinced them about state's potential and opened up new prospects of tyre grade sheet in Tripura. Under the banner of National Institute for Rubber Training, Kottayam there are provisions for skilling and up-skilling of latex processors.

3.3 Sheet Making Setups

Sheet making units are basically of three types, the majority being in producers' backyards with bare minimum processing facilities with or without smoking setups. Rests are with designated Group Processing Centres (GPCs) belonging to RPSs, State-run-corporations (PSUs) like TFDPC Ltd. and TRPC Ltd. and with few latex aggregators like CPUs, private enterprises, dealers and processors. Most of the sheet-making setups, having being with individual producers, do not have required infrastructure and potential for RSS making. This is mainly attributed by the absence of smoking facilities. Thus 30% of the total production of sheet rubber, produced in Tripura, is still of off-grade (ADS).

Individual latex processing and sheet making units cover less than or about 1 ha of plantation on an average. Private enterprises, CPUs, RPSs, PSUs and few individual producers have larger catchment areas. GPCs of private enterprises, RPSs, PSUs and CPUs have average capacity to handle production of about 75 ha. Due to market interventions GPCs in RPS & Corporation sectors have leaned towards supply of PFL which delivers higher rates than RSS grades. Recently Tribal welfare Dept., Govt of Tripura has taken up initiatives for bringing out small scall smoke houses across the state. There are about 14,000 individual setups together with 200 Group Processing Centres and 150 entrepreneurs that are involved in sheet rubber production.

3.4 ISNR Making Units

ISNR manufacturing setups are mostly situated in and around Rubber Park, Bodhjungnagar Industrial Area. Field Coagulum (FC) apparently has abnormally higher share in the total natural rubber production in the state. This is attributed by production of rubber, mainly scrap rubber from unaccounted plantations raised in undocumented areas, poorly attended plantations (by owners) and from pilferages in certain cases.

There are eight ISNR making units in Tripura out of which one in public sector other being closed down. Major grade of ISNR produced in Tripura is ISNR 20.

SI	Name of the Unit
1	TFDPC Ltd. Takmacherra, South Tripura
2	Bright Rubber Processors (P) Ltd. Bodhjungnagar, West Tripura
3	Malaya Rub-Tech Industries Rubber Park, West Tripura
4	Ramakrishna Rubber Industry Rubber Park, West Tripura
5	ABM Rubber Industries Pvt. Ltd. Rubber Park, West Tripura
6	Palappillil Techno Rubber Rubber Park, West Tripura
7	Mass Latex Rubber Park, West Tripura
8	Lakshya Polymar Industries Rubber Park, West Tripura
9	Hima Rubbers Pvt. Ltd. (upcoming) Bodhjungnagar, West Tripura
10	Alwaye Rubber Pvt. Ltd. (upcoming) Rubber Park, West Tripura
11	Agartala Rubber Industries (Closed) Bodhjungnagar, West Tripura

3.5 Cenex Making Units

There are four Cenex making units in Tripura out of which one in public sector and rest three with private sector. One Cenex factory is in Bodhjungnagar Industrial Area with the rest 3 outside of it.

SI	Name of the Unit
1	TFDPC Ltd. Takmacherra, South Tripura
2	Season Rubbers Private Ltd. Gabardi, West Tripura
3	Trumaxx Rubber Private Ltd. C/o Abhisar Buildwell Private Ltd. Bodhjungnagar, West Tripura
4	Virgin Private Ltd. (Closed) Simna, West Tripura

3.6 Rubber-wood Processing

There are only three rubber wood processing units in Tripura, two belonging to TFDPC Ltd. & other is privately owned Treatment Plant.

SI	Name of the Industry
1	TFDPC Ltd. Anandanagar, West Tripura
2	TFDPC Ltd. Betcherra, Unakoti Tripura
3	Laxmi Wooden Treatment Plant Barjala, West Tripura

The units, owned by TFDPC Ltd. do not entertain wood from individual rubber growers as its installed capacity is even inadequate for own consumption. Rubber producers go harvesting in their senile plantations as there is no remunerative price realization against sale of rubber trees on felling for replanting.

Three rubber wood-based plywood factories are coming up during next financial year.

4. Local Consumption for Product Development

4.1 Cenex

SI	Name of the Industry	Product
1	Abhisar Buildwell Private Ltd. Bodhjunnagar, West Tripura	Thread Rubber
2	Aristo Texcon Rubber Park, West Tripura	Thread Rubber
3	Maximus Rubber Industries Pvt. Ltd. Rubber Park, West Tripura	Tyre Retreading Rubber

Cenex is the only produce that gets domestically consumed as raw material to almost full extend.

4.2 Local Consumption Scenario

Local consumers of NR in Tripura are primarily latex based. They consume less than 4% of total rubber production of the state.

5. Ancillary Income Generation

5.1 Intercropping

Rubber plantations provide rich sources of space, nutrition and sunlight during first 3-4 year from planting to cultivate a good number of annual crops as intercrops. Upland paddy, banana and pineapple are most common intercrops in rubber plantations in Tripura. There still persists prospects of raising other annual crops like ginger, turmeric, sesamum, leafy vegetables by arranging irrigation or water harvesting facilities.

Another set of shed loving plants can also be raised as intercrops in mature plantations. Last year Sugandhimantri (an aromatic medicinal plants) has been raised in few hectares of mature areas on trial basis.

5.2 Rubber Honey

Rubber tree is a rich source of honey. As rubber honey is available in petiole region, honeybees can access almost all the honey available with the rubber tree. One hectare of rubber plantation can accommodate 10 beehives with potential of 8 to 10 kg rubber honey annually. There are about 139 ha rubber plantation in Khowai district which is harvested for rubber honey in an organized manner. Honeybees elsewhere in the state also access rubber plantations for collection of honey.

5.3 Rubber seed

Rubber seed in Tripura is solely used for multiplication in Tripura. Tripura has been the rubber nursery hub in entire North East India. Tripura is now self-sufficient in meeting its domestic demand of planting materials for new/re-planting. Besides meeting domestic demands, state has been supplying lakhs of planting materials to Assam, Manipur and Mizoram since 2021 for implementation of INROAD scheme aiming at raising 2 lakh rubber plantations during 2021-25. Rubber seed cake is rich source of protein which can be used as a good fodder. Rubber seed oil is used for soap, paint and resin industries. Such prospects are still untapped in the state. Being the first time in history, Tripura has supplied 12 mt assorted rubber seed to Kerala this year.

6. Marketing Procedure

6.1 Moods & Modes

Rubber market fluctuates depending on a number of national & international trends that even fluctuate rates of petroleum, gold, silver etc. on minute-to-minute basis. So, it is evident that rubber growers might end up with a high, as well as a very low sales realization despite their cost of cultivation remains the same in a year and certainly increases over the years. Still rubber marketing in Tripura is the most spectacular in regard to its strengths, spirits, spreads, swings, sentiments and successes. Advance payment to the producers still remains an important tool in customer relationship management. This relationship depends on market sentiments. At times very close and intimate dealer stops lifting phone. This may even happen reverse way when producer disconnects after availing advance payment from the dealer. Surprisingly in this era of digital dealings, many grocery shops in rural areas still run barter economy by exchanging rubber with eatables and household commodities.

The phenomenal success in growth of rubber development in Tripura has been dependent on many factors of which available marketing facilities of the produce is the prime one among them. As on date there are 1,636 rubber dealers in Tripura producing only about 10% of total domestic rubber production whereas Kerala, producing about 83% of total production has only 5,331 rubber dealers. Eight very good running ISNR factories consume almost 100% of field coagulum, produced in the state. Three Cenex factories of the state consume entire quantity of PFL, produced in the state.

The Rubber Board has launched electronic platform, named as mRube for online dealing in 2022-23 which has enhanced visibility of entire rubber market scenario through a palmtop or desktop and allows rubber dealers dealing their rubber at the best price across the country, sitting in Tripura. mRube provides rooms for price discovery, bargaining and interest free advance payment provision from Escrow account. One registered mobile number and an email account are sufficient to join this digital marketing portal. Manimalayar Rubbers, Agartala Branch has dealt about 23% of total mRube turnover during 2022-23.

Rubber Board promoted Manimalayar Rubbers Pvt. Ltd. renders all sorts of marketing support to rubber growers, producers and other stakeholders. Declared price of Manimalayar becomes the bench mark price for rubber sheet in the state. Manimalayar Rubbers has even constituted another trading company, Tripura Latex Pvt. Ltd. to get involved in marketing in rubber sector. Rubber Producers' Society (RPS) is yet another support system that delivers excellent handholding backup to small producers by collective processing and marketing. Other aspects include licensing for dealing, processing, manufacturing endeavours and publication of rubber rates to local dailies.

6.2 Destinations of the Produce

Most of the sheet rubber is sent to North India, some quantity even to West & South India. Non-tyre sector had monopoly in Tripura rubber market until 2017-18. Still today large dealers are common customers while tyre grade sheet is also sent to giant tyre manufacturers, like MRF, Apollo, J K Tyre, CEAT etc.

ISNR and Creep go mainly to North India. They are mostly used in sectors like tyre, agriculture implements, spare parts etc.

Almost total quantity of Cenex, produced in the state, got consumed locally mainly for rubber thread manufacturing.

Rubber thread is mainly marketed to North India, rest to West and East India.

7. Overall Activities of Rubber Board at Agartala

7.1 Rubber Plantation Department:

Rubber Plantation Department is the largest department in the Rubber Board and involved in planting material generation, plantation development, harvesting, processing, value addition and marketing. It collects database on different aspects of rubber plantation in Tripura. Rubber Plantation Department works through the network of its six Regional Offices and twenty-five Field Stations under NRETC/Zonal Office, Agartala. All types of technical supports for plantation development, estate management, harvesting, processing etc. are made available for stakeholders on call. Financial supports are also given as plantation development subsidies, stipend & utilities for training, labour welfare, processing and smoking infrastructure development through Board's six Regional Offices. It also imparts different training programs.

There is even one setup for rendering trading including export promotion of Marketing Division, Rubber Board attached to NRETC/ZO, Agartala.

Manimalayar Rubbers Pvt. Ltd. and Tripura Latex Pvt. Ltd., promoted by Rubber Board and managed by Board's Technical Officers provide marketing support together with supply of estate inputs, machineries, chemicals etc. to rubber stakeholders.

7.2 Rubber Research:

Regional Rubber Research conducts different studies and location specific trails to recommend suitable measures for plantation development, ancillary income etc.

- Focus of rubber research in Tripura:
 - ✓ Genetic improvement in terms of breeding and selection of location-specific high yielding clones.
 - ✓ Formulation of location specific agro-management practices, nutrition management and planting techniques. This is a continuous process as new challenges emerge such as soil degradation, climate change, new pests/diseases, environmental concerns etc.
 - ✓ Latex harvesting techniques.
 - ✓ Development of rubber-based cropping systems.
- Planting material recommendations have been formulated and regularly updated.
- Ecological impact studies are conducted.
- Development of cropping systems with Banana, Pineapple, Ginger, Turmeric, Pigeon pea, Sesame, Upland rice, Colocasia, Amorphophallus, Maize and Groundnut were evaluated as intercropped in immature rubber plantations. Other crops like Tea, Cowpea, Ladies finger and Amaranthus were also evaluated during the initial years.
- Integration of socio-economic and ecological factors to develop rubber based agro-forestry systems in a homestead approach acceptable to local growers.
- Making rubber cultivation more profitable minimising the input costs and maximising rubber productivity.
- Socio-economic impact studies.
- RRII has used satellite-based remote sensing and GIS techniques in collaboration with ISRO, Bangalore and IIT, Kharagpur for spatial mapping of existing rubber plantations in Tripura.
- In order to prepare soil fertility map for rubber growing regions of NE India and to develop an on-line fertiliser recommendation (Rubber Soil Information System (RubSIS), collection of geo-referenced composite soil samples per every 40 ha of rubber holdings was initiated.
- Planting of Genetically Modified (GM) rubber in 2024.

7.3 Regional Rubber Training Institute:

Regional Rubber Training Institute imparts training in different human resource development programs of National Institute for Rubber Training (NIRT), Kottayam, Kerala. Agartala Nodal Centre of NIRT has been opened up in 2022-23. This institute has conducted Post Graduate Diploma in Rubber Plantation Management for 21 students. This institute has imparted training to five B. Voc. students of Tripura University this year. Recently it has also imparted induction training to newly-recruited 34 Field Officers of NE Region. Routinely this institute organizes all rubber-based skilling and up-skilling programs for growers, tappers, processors, entrepreneurs through Board's Regional Offices.

8. Green Nature of Rubber

Rubber cultivation is very environment friendly and leaves a good number of positive impacts on reducing global warming & many other environmental menaces.

- 8.1 Other option with synthetic rubber involves 108-174 GJ per MT energy for production whereas 15-16 GJ per MT energy is required for production of natural rubber. Most of this energy comes from solar power in case of natural rubber whereas electrical energy is mostly utilised for production of synthetic rubber.
- 8.2 The rates of photosynthesis in rubber plantations is recorded as 10-15 $\mu\text{mol CO}_2$ per m^2 per second against 5-13 $\mu\text{mol CO}_2$ per m^2 per second in natural vegetations.
- 8.3 Carbon sequestration capacity of rubber plantation is also significantly higher (7.80 t C/ha/yr) compared to much less figures of Tropical South America Rainforest, Pine Forest (USA), Successional Temperate Deciduous Forest (USA), Various Temperate Forests, Tundras, US Forests & Woodlands, European Forests, Amazon (Rondonia) forests.
- 8.4 Soils in rubber plantations have been found to be richer with organic matter and microbial populations compared to teak & other plantations.
- 8.5 Rubber plantations reduce pressure on natural forest for timber as rubber wood is as good as teak & other conventional woods.

9. People Involved

In the backend of rubber sector there are nursery owners who engage general duty workers and skilled budders for budding. In plantation sector rubber growers engage general duty workers together with skilled tappers and processors. In case of large processing units producing ISNR and Cenex, a large number of people get engaged in different job-roles. In manufacturing sector also there are different sets of manpower involved. Besides these, rubber dealers and logistic personnel are also involved in forward linkages.

10. Conclusion

Rubber has been the backbone of Tripura economy for decades together. Besides state's economy, this crop shapes entire social panorama. Rubber in economic front arranges the highest revenue for the state.

SI	Particulars	2019-20	2020-21	2021-22	2022-23
1	Taxpayers in rubber sector (No.)	6,226	7,285	8,573	7,904
2	Assessable value (Rs. Cr.)	1,280	1,700	2,240	1,620
3	Production (mt)	83,701	90,712	93,371	1,02,989
4	Unit price (Rs./Kg)	120	140	181	139
5	GST Collection (Rs. Cr.)	64	85	112	81